CONNOR REYNOLDS

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A cunning computational biologist experienced in the application of genomics large language models for genomics research and single-cell data analysis.

EMPLOYMENT

Postdoctoral Research Assistant, Earlham Institute — **03.2026 – Present**

- Funded through an intitute development grant to innovate genomics large language model (gLLM) applications for genomics research in plants.
- Involved first ML-driven evolutionary constraint mapping of the *Arabidopsis* and potato genomes using **zero-shot masked imputation scores** in the gLLM model **PlantCAD2.**
- Sucessfully applied for a compute grant awarding access to **Isambard-AI supercomputing resources.**
- Included classification and interpretation of highly- and lowly-variable-genes based on single cell expression in Arabidopsis by fine-tuning PlantCAD2 with a test performance F1 score of 0.84.
- Identification of important motifs and k-mers driving gLLM predictions using gradient-activation.

Genomics Data Scientist, TraitSeq Ltd (part-time during PhD**)** — **09.2024 – 02.2026**

- Provided data science support within an AgTech start-up company focused on predicting phenotyptic traits for precision breeding and crop health.
- Identified gene expression biomarkers displaying consistent significant associations with herbicide response based on 5 phytoxicity measurements related to plant health & fitness in maize field data.
- Supported the developmet of agentic pipelines for bioinformatics analyses of multiomics data.

Alan Turing Institute Research Fellow, Earlham Institute / Alan Turing Insitute — **02.2021 – 09.2022**

- BBSRC-funded data science fellowship in colloboration with The Alan Turing Institute and Norwich Bioscience Institutes.
- Embedded in the Plant Genomics group to apply techniques in machine learning and artificial intelligence for characterization of circadian clock function in plants.
- Conception of ChronoGauge – an AI-based method for estimating the circadian time in plants using gene expression data.
- Advanced training and mentorship in machine learning and data science delivered by the Alan Turing Institute.

EDUCATION

PhD in Biological Sciences, Earlham Institute (computational-based**)** — **10.2022 – 03.2026**

- Worked within the Anthony Hall Group (Earlham Institute) and Antony Dodd Group (John Innes Centre).
- Fully-funded studentship delivered by the prestigious Norwich Research Park Doctoral Training Programme.
- Thesis submitted **6 months early** in 3 years and 6 months.
- **Published 1st Author Nature Communications paper** in 3rd year (*Reynolds et al.* 2025).

Thesis title: Using machine learning to predict the circadian time of plants using gene expression

MSc by Research in Biomolecular Science, University of East Anglia — **10.2019 - 09.2020**

BSc (Honours) Biomedicine, University of East Anglia — **10.2016 - 07.2019**

INTERNSHIPS

Professional Internship Placement, Science & Technology Facilities Council (STFC) — **03.2025 – 06.2025**

- Embedded within the Machine Learning and Molecular Dynamics groups to improve the accuracy and efficiency of molecualr simulation analyses.
- Generated 1ms molecular dynamics simulation trajectories using GROMACS in lysozyme crystal structures.
- Implemented graph transformers in PyTorch to classify residue-level neighbourhoods with an F1 score of 0.81.

Bioinformatics Summer Internship, University of East Anglia — **06.2018 – 09.2018**

- Processing and sequence alignment of degradome-seq data in *A. thaliana* using the Patman short-sequence aligment tool.
- Development of a Python-based software tool for analysis and visualization of sites of degraded mRNA.
- Improvement of *in silico* techniques for predicting mRNA sites targeted by microRNA mediated cleavage

TECHNICAL SKILLS

PROGRAMMING / SCRIPTING LANGUAGES: Python, R, Bash

MACHINE LEARNING: developing and training models including multi-model ensembles and foundation models
Scikit-learn | XGBoost | TensorFlow | PyTorch | MLPs | CNNs | RNNs | Transformers | Foundation models

LARGE LANGUAGE MODELS: Agentic system development, coding accelaration and word embedding analyses
Claude | GPT | llama | DistilBERT | MiniLLM

GENOMICS LARGE LANGUAGE MODELS: fine-tuning of pre-trained gLMs for DNA prediction tasks and zero-shot evolutionary constraint prediction
HuggingFace | AgroNT | PlantCaduceus | Evo2

BIOINFORMATICS: Quality control, adaptor trimming, sequence alignment, read quanitification, expression matrix validation, differential expression analysis, high performance computing, genome-wide-association, singularity containers
Slurm | STAR | EdgeR | limma | GAPIT | Github | Agentic pipeline development

SINGLE CELL DATA ANALYSIS: Processing single-cell data, filtering of low quality cells, doublet detection, anchor identificaiton, intergration across batches, clustering , marker detection, transfer labelling, reference cell-type annotation
Cellranger | Seurat | Scanpy | DoubletFinder | Single-cell foundation model

GRANTS & AWARDS

UK NATIONAL ISAMBARD-AI ACCESS COMPUTE GRANT — **12.04.2026**
Compute grant including access to Isambard-AI GPU resources as a co-principal investigator for predicting evolutionary constraint across plant genomes as a research project

AI IN RNA BIOLOGY – RNA SOCIETY CONFERENCE AWARD — **28.09.2025**
Received Oral Presentation Award

NORWICH RESEARCH PARK PHD SUMMER CONFERENCE 2025 PRIZE — **17.06.2025**
Awarded 1st place prize for best poster

SELECTED CONFERENCE & ENGAGEMENT ACTIVITIES

PLANT & ANIMALS GENOME CONFERENCE 2026 – SAN DIEGO — **10.01.2026**
Invited speaker at Machine Learning and Systems Biology session

PLANT & ANIMALS GENOME CONFERENCE 2023 – SAN DIEGO — **12.01.2023**
Invited speaker at AI in Biology session

ALAN TURING INSTITUTE & NORWICH BIOSCIENCE INSTITUTES SYMPOSIUM — **07.07.2022**
Main speaker

2021 EARLHAM INSTITUTE INNOVATE CONFERENCE — **17.11.2021**
Invited speaker for AI in Plant Sciences session

2021 EARLHAM INSTITUTE SCIENTIFIC ADVISORY BOARD — **19.10.2021**
Invited speaker at Genomics for Food Security session

PUBLICATIONS

Cano-Ramirez D† **Reynolds C†,**Thomas P, Hall A and Locke JCW. Cell-to-cell transcriptional variability is structured across cell types in the *Arabidopsis* root.
(† denotes equal contribution, manuscript in preperation for submission)

Goswami R, Cano-Ramirez D, Abley K, Afsharinafar M, **Reynolds C**, Zeng J, Bhattarai M, Zhang F, Cortijo S, Hall A, Sandro-Colizzi E and Locke JCW. Regulatory architecture tunes gene expression noise in the plant heat shock response.
(Manuscript in preperation for submission)

Reynolds C, Colmer J, Rees H *et al.* Machine learning models highlight environmental and genetic factors associated with the *Arabidopsis* circadian clock. **Nature Communications** 16, 7223 (2025). https://doi.org/10.1038/s41467-025-62196-w

Rees H, Rusholme-Pilcher R, Bailey P, Colmer J, White B, **Reynolds C**, Ward SJ, Coombes B, Graham CA, Dantas LLB, Dodd A and Hall A. (2022) Circadian regulation of the transcriptome in a complex polyploid crop. **PLOS Biology** 20(10): e3001802. https://doi.org/10.1371/journal.pbio.3001802